Abheek Dhawan

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EDUCATION

- **Yale University** New Haven, CT
  B.S. Computer Science & Astrophysics | GPA: 3.92 *August 2024 – May 2028*
  Relevant Coursework: *Algorithms, Data Structures, Systems Programming, Database Design & Implementation, Machine Learning, Astrostatistics & Data Mining, Language and Computation I, Creative Embedded Systems, Quantum Mechanics*

EXPERIENCE

- **CourseTable** New Haven, CT
  Software Engineer *August 2024 – Present*
  - **Product Development**: Contributed to development and maintenance of CourseTable, Yale's course-planning platform used by more than 95% of students; improved UI/UX and course search in an established TypeScript/React codebase.
  - **Quality & Reliability**: Contributed testing and bug fixes, and implemented PWA installation and offline caching to support access to cached course-planning content without a network connection.
- **Universidad de Chile** Santiago, Chile
  Astrophysics Research Assistant (Full-time) *June 2026 – August 2026*
  - **Scientific Analysis**: Built a modular Python/NumPy/pandas pipeline for 1,380 SPHINX20 galaxies across high-redshift snapshots, enabling systematic comparison of star formation over 3–100 Myr timescales.
  - **Research Validation**: Implemented reusable regression and bootstrap analyses to quantify main-sequence slope, scatter, and short-timescale variability; reproduced published results and validated conclusions across two independent simulations.
- **Yale Quantum Institute** New Haven, CT
  Software Engineer (Full-time) *June 2025 – August 2025*
  - **Deployed Courseware**: Built and deployed *visual-notes* in React/TypeScript/Three.js for Yale CPSC 4470 (about 50 students); used in lectures and supplementary material to teach 7 quantum-computing concepts.
  - **Interactive Graphics**: Developed reusable, scroll-driven circuit, Bloch-sphere, and Feynman-path visualizations with React Three Fiber, Framer Motion, and Scrollama; established Jest, ESLint, and Husky testing/linting workflows.
- **St. Louis Modern Chinese School** St. Louis, MO
  Programming Teacher (Part-time) *August 2023 – May 2024*
  - **Programming Instruction**: Taught 50 students weekly lessons in introductory programming, Python classes/OOP, and web development through the Coding Academy St. Louis nonprofit.

PROJECTS

- **Wwise-Audio-Tools** — *C++, CMake, Binary Formats* github.com/WolvenKit/wwise-audio-tools
  - Reverse-engineered Wwise audio formats and built reusable C++ libraries and a CLI with CMake to decode game audio assets from WEM to OGG.
  - Integrated as a shared dependency in WolvenKit (1.2k+ GitHub stars), enabling the modding toolkit to decode game audio programmatically as part of its asset-processing workflows.
  - Developed additional game mods that garnered 15,000+ downloads on Nexus Mods.
- **abheekd-lang** — *C++, LLVM, Compilers* github.com/abheekda1/abheekd-lang
  - Built an end-to-end C++ compiler for a custom language, implementing a handwritten lexer/parser, LLVM IR generation, and native object-code compilation.
- **AwesomeSciBo** — *TypeScript, Node.js, Docker* github.com/abheekda1/awesomescibo
  - Built and maintained an open-source Science Bowl bot serving more than 2,000 peak users; deployed with Docker to support competitive Science Bowl practice.
  - Implemented targeted practice rounds, slash commands, and cross-server leaderboards; maintained the bot as training software for Yale Undergraduate Science Bowl.
- **Sonara** — *Python, NLP, Speech-to-Text* github.com/abheekda1/sonara
  - Integrated Deepgram speech-to-text and Hugging Face zero-shot classification to route patient tasks from clinical transcripts, helping reduce clinician note-taking time by 35%; 2025 NYC ASA AI4Purpose Hackathon Runner-Up.

SKILLS

Languages: Python, C++, C, Go, TypeScript/JavaScript, Java, SQL, Bash, Assembly
Web & Graphics: React, Next.js, Node.js, FastAPI, Flask, REST APIs, gRPC, PWAs, Three.js, React Three Fiber
Data & ML: PyTorch, scikit-learn, NumPy, pandas, SciPy, Hugging Face, spaCy, Astropy; regression, bootstrapping
Systems: CMake, LLVM, GDB, Valgrind, multithreading, networking, async I/O, binary formats
Cloud & Databases: Linux, Docker, Kubernetes, AWS, PostgreSQL, MongoDB, Redis
Testing & Delivery: Git, GitHub Actions, CI/CD, Jest, ESLint, Husky

LEADERSHIP & AWARDS

- **President, Yale Undergraduate Science Bowl**: Hosted a Science Bowl competition for over 120 students; maintain the club's AwesomeSciBo software to support team training.
- **Captain, Yale Jashan Bhangra**: Lead Yale's Punjabi folk dance team in collegiate competitions across the U.S.
- **Honors**: Tetelman Fellowship (2026); NYC ASA AI4Purpose Hackathon Runner-Up (2025); 21st in U.S., USAAAO National Astronomy Competition (2023).